COSC2148/COSC3130 Computing Research and Project Preparation

Assignment 2

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Section 1

Introduction

1.1 DELETE original question

Introduction - 1 page (2 marks)

Briefly describe your topic and why it is important or interesting. You could start by describing the area of interest and later summarise the existing solutions and their limitations. You could also explain the specific focus of your research and why you are doing so. Feel free to include specific information, such as detailed statistics, that will back up your claims (and thus make your statements much stronger).

In summary, an introduction should:

- Introduce the topic and its significance.
- Give background and context.
- Outline your problem and research questions.

Some important questions to guide your introduction include:

- Who is interested in the topic (e.g. scientists, practitioners, policymakers, particular members of society)?
- How much is already known about the problem?
- What is missing from current body knowledge?
- What new insights will your research contribute?
- Why is this research worth doing

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Section 2

Research Questions

2.1 DELETE original question

Research questions - 1 page (4 marks)

This section should address the technical issues you will be addressing in your research. It is about the "what" (you will be addressing/focussing when you start your thesis).

Describe here the specific research questions to be addressed. You need to break the problem you are addressing into specific sub-problems (called here research questions) that could help you during the thesis to come up with innovative solutions properly.

Show how the research questions are interesting and worth addressing.

Please include at least two (2) research questions, but you are limited to a maximum of three (3) research questions.

For each research question, describe what it is, its significance, and why it is important to address it.

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2.2.1 aaaSection2Subsection1

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Section 3

Literature Review

3.1 Overview of Fake News Detection

- Introduce fake news and inauthentic content problem
- Explain importance of detection in social media
- Evolution of approaches:
  - Text-only models
  - Context-aware models
  - Multimodal models
- Content alone is not sufficient, context and behaviour matter
- Deep learning dominates modern approaches
- Different approaches address different aspects of the problem

3.2 Content-Based Approaches

- Models analyse only textual features
- Techniques:
  - NLP, BERT, RNN, CNN
- Examples:
  - Text branch in DANES and GETAE
- Strengths:
  - Simple and efficient
  - Good baseline performance
- Weaknesses:

  - Cannot detect visual misinformation
  - Ignores user behaviour and propagation
  - Struggles with subtle misinformation
- Content-only approaches are not sufficient for real-world scenarios

3.3 Context-Aware Approaches

- Use social media context instead of only content
- Types of context:
  - User behaviour
  - Interactions
  - Propagation patterns
- Models:
  - DANES (text + social context)
  - GETAE (text + propagation graph)
  - GCAN (context only)
- Techniques:
  - Graph neural networks
  - Propagation graphs
- Strengths:
  - Captures real-world behaviour
  - Improves detection performance
- Weaknesses:
  - Relies on text-only content
  - Ignores multimodal data
  - Depends on specific datasets
- Context improves detection but content is still limited

3.4 Multimodal Approaches

- Combine multiple data types such as text and images
- Social media content is inherently multimodal
- Models:

  - EANN, MCNN, HMCAN, MPFN, LDPG

- Techniques:
  - CNN, transformers, fusion methods
- Strengths:
  - Better feature representation
  - Improved accuracy
  - Detects misleading visuals
- Weaknesses:
  - Ignores social context
  - Static models without propagation
  - Requires large datasets
- Multimodal improves content but lacks context integration

3.5 Research Gap and Positioning

- Content-based models are limited
- Context-aware models rely on text-only content
- Multimodal models ignore social context
- There is no effective integration of multimodal and context-aware approaches
- Existing models such as DANES and GETAE use text-only content branches
- Proposed research replaces text branch with multimodal branch
- Combines multimodal content with context-aware architecture
- Evaluates performance improvement and contribution of each branch

Section 4

Research Methodologies

This research adopts a comparative experimental design to investigate the effectiveness of multimodal content and social context features in the detection of inauthentic social media posts.

note Dellys et al. (2025) is flagged as the closest prior work (how does this differentiate?)

4.1 System Overview

Prior research has demonstrated that combining text-based content features with social context improved detection performance in identifying inauthentic social media posts. Separate studies have shown that multimodal models can outperform text-based models.

Limited research has been conducted to investigate the combined effectiveness of multimodal content features and social context features by substituting text-based representations with multimodal representations within a context-aware framework.

The proposed system aims to address this gap by utilising existing DANES and GETAE feature ensemble architectures that combine text content with social context features, and extending them to incorporate multimodal content features.

4.2 Proposed Framework Architecture

4.2.1 Baseline Architecture: DANES/GETAE

DANES (citE) and GETAE (cite) each comprise two branches: a text-based content branch that encodes textual content using word embeddings and transformer-based representations in a neural network, and a social context branch (referred to as the Propagation Branch in GETAE) that models the social network user interactions and behaviour using graph neural networks and deriving node embeddings.

Outputs from these branches are combined using an ensemble approach produce a classification embedding. These models achieve state-of-the-art performance in detecting inauthentic social media posts in Twitter15 and Twitter16 datasets, outperforming text-only models and other context-aware models that do not utilise multimodal content features.

These models represent a principled starting point for the proposed architecture modification.

4.2.2 Proposed Modifications to Baseline Architecture

The proposed architecture modification involves substituting the text-based content branch with a multimodal content branch that incorporates visual features extracted from social media posts as well as textual features. The social context branch is retained, as in the baseline architectures, allowing the contribution of the multimodal content branch to be evaluated against the baseline models.

4.2.3 Multimodal Branch Design

This research will adopt a well validated multimodal architecture from the literature that has been designed for the task of multimodal fake news detection. This approach maintains a focus on the architectural substitution question rather than the design and optimisation of a novel multimodal architecture.

The key considerations for the selection are: task alignment, dataset compatibility, ease of integration and widespread validation.

From our review of the literature, HMCAN and MPFN models have both demonstrated excellent accuracy (0.89.7% and 83.3% respectively) in multimodal fake news detection using the Multimodal Twitter dataset, outperforming other common multimodal models for fake news detection (such as EANN). HMCAN's BERT + ResNet encoding pipeline is consistent with the text branch of DANES/GETAE and has been independently validated on the Multimodal Twitter dataset as well as Fakedit dataset, making it a suitable candidate for the multimodal content branch. MPFN utilises a more complex image pipeline which could be beneficial for detecting manipulated or artificially generated images, and is another potential candidate.

Our system will adopt HMCAN with MPFN as a fallback option as this provides a robust foundation for handling multimodal content.

4.2.4 Experimental Design

The initial stage is to replicate the published DANES/GETAE architectures and evaluate their performance on the Twitter15 and Twitter16 datasets and ensure that the results are consistent with those reported in the original papers.

This is followed by the transition to a dataset that contains multimodal content and the implementation of the proposed architecture modification to incorporate multimodal content features.

The following configurations are proposed for evaluation:

- **Stage 1 — Condition A (Implementation Baseline):** Replicate the published DANES/GETAE architectures using the Twitter 15/16 datasets to confirm the implementation reproduces performance consistent with the original papers before any modifications are introduced.
- **Stage 2 — Condition A (Dataset Baseline):** Evaluate the DANES/GETAE architectures on a multimodal dataset in text-only mode (i.e. ignoring the visual content)

to establish a baseline for the DANES/GETAE architectures on the new dataset that Stage 2 conditions will be compared with.

- **Stage 2 — Condition B (Ablation 1):** Text-only content branch with no social context branch to isolate the contribution of textual features without social context features.
- **Stage 2 — Condition C (Ablation 2):** Multimodal content branch with no social context branch to isolate the contribution of multimodal features without social context features.
- **Stage 2 — Condition D (Proposed):** Multimodal content branch with a social context branch to represent the full proposed framework for primary evaluation.

Each stage 2 condition is compared against the stage 2 condition A baseline to evaluate the attribution of performance improvements.

RQ1 is directly addressed by comparing the performance of the proposed architecture (Condition D) against the baseline architecture (Condition A) with the multimodal dataset.

A pairwise comparison of condition's B, C and D address RQ2 by evaluating the contribution of the multimodal content branch and the social context branch to the overall performance of the model.

It is anticipated that the transition from Twitter15/16 datasets to a multimodal datasets will result in a performance difference in Stage 2 Condition A relative to Stage 1 depending on the extent to which these models generalise to a new dataset. This does not impact the evaluation of the proposed architecture change since all Stage 2 conditions are evaluated against the Stage 2 Condition A baseline. (TODO: this would require a mention under the LIMITATIONS section, we aren't interested in researching the generalisability of these models this is rather an expected degradation)

This methodology enables direct evaluation of the research questions by isolating the impact of multimodal content through controlled substitution of the content branch and by analysing the relative contribution of content and context features through ablation experiments. The structured experimental design ensures that performance differences can be attributed to specific components of the model.

4.2.5 Processing Pipeline

4.2.6 Evaluation Metrics

4.2.7 Datasets and Data Preparation

4.2.8 Timeline

Section 5

Evaluation

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evaluation - 1 page (3 marks)

This section is important and should clearly show the evaluation plan of your research. This evaluation plan could be used later during your thesis to clearly demonstrate that you made research contributions in the field of study.

This section will need to address the following points:

- Benchmarking: clearly select some existing methods/algorithms you will need to compare your research work with. It is important to clearly explain these methods/algorithms as well as justify the reasons for choosing them.
- Data sets: describe at least two well-accepted data sets that you will be using to compare your research work against the selected benchmarked methods/algorithms. Again, you will need to justify the reasons for selecting such data sets.
- Metrics: to compare your research work with benchmarked methods/algorithms, you clearly need to list the metrics to be used for this purpose. This could be, for example, "execution time" if you are comparing the performance of your work with other methods/algorithms. You also need to explain why you have selected such metrics. List at least two metrics.

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Section 6

Project Management

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project management - 1 page (1 mark)

This section should outline how you plan to manage your research project throughout the semester.

It is essentially a Project Plan that clearly defines the scope and deliverables, including explicit exclusions.

It should present a structured schedule with milestones and a transparent critical path, while assigning responsibility to specific 2individuals to ensure accountability.

It must also incorporate risk management to anticipate threats and formulate appropriate responses, protecting project objectives from avoidable failure.

This section should also clearly state the methods you will use to manage the project, such as Agile sprints, Scrum ceremonies, Kanban, etc., and specify the enabling tools (e.g., MS Project, Trello, JIRA, Slack, etc.) that will operationalise these methods.

Briefly explain how these methods and the supporting tools together will facilitate scheduling, progress tracking, communication, and reporting. Specifically, you must include the following:

- High-level scope and deliverables of your research project. Clearly state exactly what you will deliver and explicitly rule out what you will not.
- Timeline with major milestones derived from the Work Breakdown Structure (WBS). This should reflect the schedule of your research project and include any critical paths. Note: you don't need to include the WBS itself.
- Clear assignment (ownership) of each task to project team member(s), not roles, for accountability and monitoring.
- Approach in risk management, identify risks and proposed mitigation and contingency plan, as well as escalation paths. List the major risks, rank them, and state what you will do before, when, and after they occur.

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Bibliography

Appendix A

Generative AI Usage

Declaration

[REPLACE THIS LINE WITH YOUR GENERATED DECLARATION FROM `https://aiattribution.github.io/`]

Tool Report: [Tool Name, e.g. ChatGPT / RMIT Val / Copilot]

Field	**Details**
1. GenAI tool used	[e.g. ChatGPT 4o, RMIT Val, GitHub Copilot]
2. Purpose for this assignment	[e.g. Brainstorming research questions, drafting outline, rewriting for clarity, code generation]
3. Prompts or instructions given	[Include key prompts used. If conversation was lengthy, include all your prompts in full and use [. . .] to truncate AI responses only.]
4. Number of prompts given	[e.g. 7]
5. Summary of GenAI output	[Describe what the AI produced. Include brief excerpts where relevant.]
6. Verification of GenAI output	[Explain how you checked the AI's output for factual accuracy, source validity, and logical soundness. e.g. "All cited sources were manually located and verified via Google Scholar."]
7. Usage or modification of output	[Explain how you used, adapted, or discarded the AI-generated content in your final submission.]
8. Reflection	[What worked well? What did not? How did it support your learning? What value did it add?]

Table A.1: Generative AI usage report — [Tool Name]